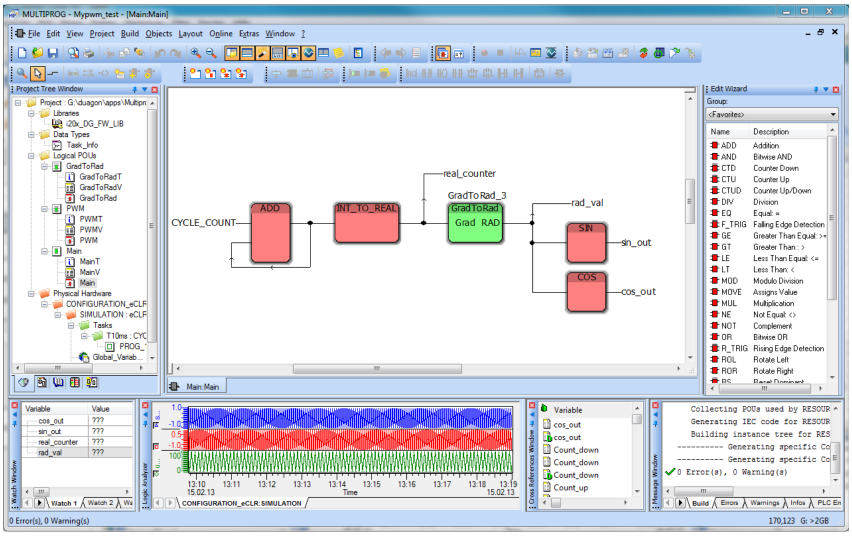
ionia IEC 61131 API

Data sheet

This document describes the API used to create ionia application programs in an IEC 61131 style development environment.



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Moving TCMS Solutions

Duagon Data Sheet Preamble

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Document history:

Rev.	Creator Date	Creator Name	Released Date	Released Name	Comments	Ident-Number
1	25. Sept. 2012	Ch. Bollinger	30.05.2013	Eberli	See chapter "Document History"	d-006935-013206
2	4. 6. 2013	Ch. Bollinger	04.06.2013	Ch.Bollinger		d-006935-014140

Table of Contents

1. Introduction

This document describes the Function Blocks (FBs) and IO-drivers that are available to program an ionia system.

1.1 Release Information

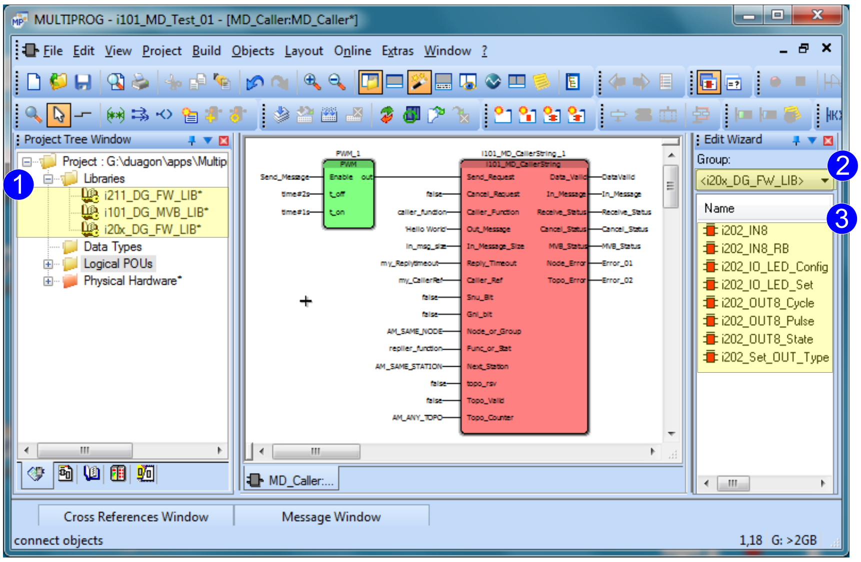
This document refers to the following versions of the components:

Deliverables	Reference
IEC61131: Installer MULTIPROG Express for ionia. Includes the libraries here below:	d-007271-014411
• i201_DG_FW_LIB R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i201_DG_FW_LIB_0010 **revision: 11323**
• i202_DG_FW_LIB R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i202_DG_FW_LIB_0010 **revision: 11324**
• i205_DG_FW_LIB R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i202_DG_FW_LIB_0010 **revision: 11327**
• i211_DG_FW_LIB R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i211_DG_FW_LIB_0010 **revision: 10457**
• i101_DG_MVB_LIB R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i211_DG_FW_LIB_0010 **revision: 11022**
• i101_DG_MVB_DRV R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i101_DG_MVB_DRV_0010 **revision: 10456**
• i101_DG_MVB_DLG R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i101_DG_MVB_DLG_0010, **revision: 11339**
• i103_DG_CO_DRV R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i101_DG_MVB_DRV_0010 **revision: 11017**
• i103_DG_CO_DLG R0010	http://svn/apps/Visual Studio/ionia_DG_Solution_Released/i101_DG_MVB_DLG_0010, **revision: 11018**

2. MULTIPROG Interface

2.1 Firmware Interface

Function (FU) and Function Blocks (FB) provide the application programming interface (API) to the various I/O modules and to non cyclic transmitted data, such as the MVB message data. The FUs and FBs are included into MULTIPROG by means of adding libraries. For each module resp. data communication type a specific library exists.



1. Libraries are included into MULTIPROG by means of a right click onto Libraries within the Project Explorer. The libraries can be stored anywhere on your local hard disk or on a nework drive.

   The default location is:
   c:\ProgramData\KW-Software\MULTIPROG Express\5_35_542\plc\FW_LIB\

2. Through the Edit Wizards filter option, it is possible to filter for a specific library.

3. According the selected filter are only the FUs and FBs shown belonging to the selected card.

2.1.1 Naming Convention

i20x_DG_FW_LIB

Card Type **duagon** **Library Type** **Library**

Card Type:

- i101 - MVB CPU Card
- i201 - Digital input card
- i202 - Digital input / output card

- i205 - Digital relay output card
- i211 - Analog Input card

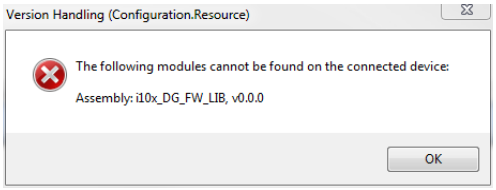
Library Type:

- FW - FU / FB Blocks
- MVB - MVB Message data driver

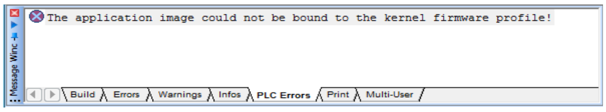
2.1.2 Versioning System

Two error cases are detected by the versioning system:

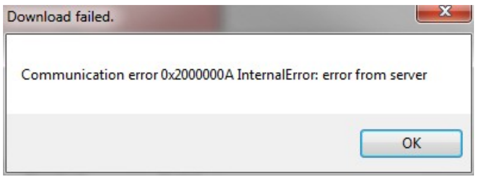
1. The application program uses a firmware library which is not available on the PLC.
   Error indication pop up window:



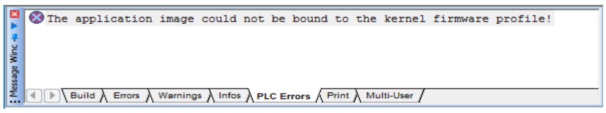
   Error indication within MULTIPROG



2. The application program uses a firmware library which is different to that running on the PLC.
   Error indication pop up window:



   Error indicaton within MULTIPROG

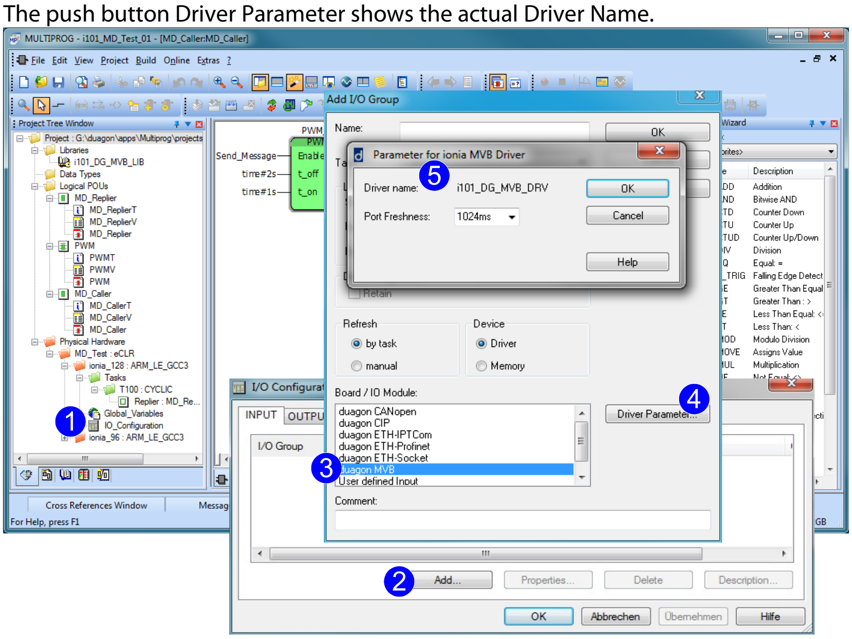


The error is revealed at download time in both cases.

2.1.3 IO-Driver Interface (Fieldbus)

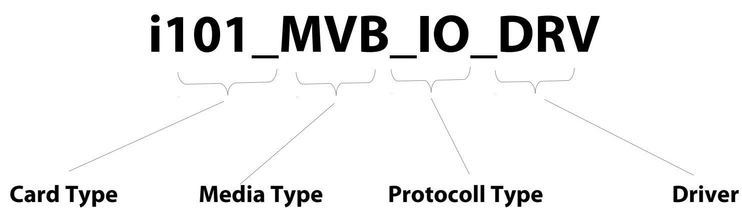
The interface to the fieldbus is provided by the so called I/O-Driver interface of MULTIPROG. The I/O configuration dialog is opened by means of a double click on the IO_Configuration icon. By pressing the push button Add at this dialog the appropriate fieldbus driver can be selected.

The push button Driver Parameter shows the actual Driver Name.



1. Double click on the icon IO_Configuration opens the I/O Configuration dialog.
2. Press the Add push button open the dialog Add I/O Group.
3. Select the appropriate field bus.
4. By pressing the push button Driver Parameter, the Parameter dialog is opened.
5. Where the actual driver name is displayed and further settings can be adjusted.

2.1.4 Naming Convention



Card Type:

- i101 - MVB CPU
- i103 - CAN CPU

Media Type:

- MVB - Multifunction Vehicle Bus

- CAN - Controller Area Network

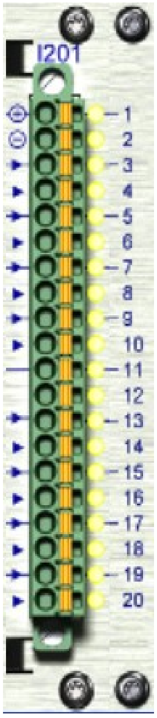
Protokoll Type:

- IO - IO Driver for Process Data Communication
- CO - CANopen 2.0 Process Data Communication

3. duagon Firmware Library

3.1 i201 Digital Input Module – i201_DG_FW_LIB Library

i201 Hardware Interface



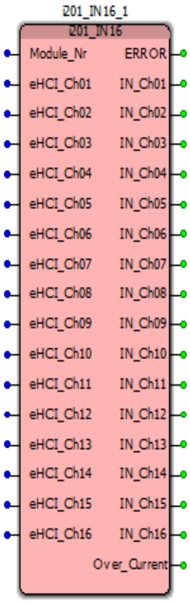
Pin	Chan	Group	Meaning
1	-		Vbat+
2	-		Vbat-
3	1	1	IN
4	2	2	IN
5	3	1	IN
6	4	2	IN
7	5	1	IN
8	6	2	IN
9	7	1	IN
10	8	2	IN
11	-		-
12	-		-
13	9	1	IN
14	10	2	IN
15	11	1	IN
16	12	2	IN
17	13	1	IN
18	14	2	IN
19	15	1	IN
20	16	2	IN

I201 Software Interface

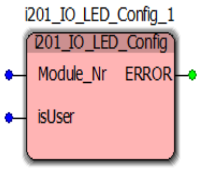
The following function blocks are included in the i201_DG_FW_LIB library

Function blocks	Purpose	Type
i201_IN16	Get digital input signals	input
i201_IO_LED_Config	Set the LEDs into user controlled mode or input controlled mode	input
i201_IO_LED_Set	Set the LEDs ON or OFF	Indication

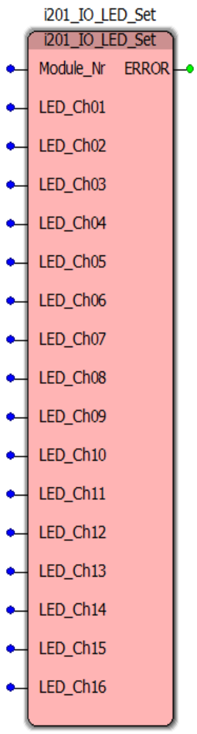
3.1.1 i201_IN16

Name	i201_IN16	
Purpose	Get digital input signals (normal inputs)	
Module Class	i201	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	eHCI_Chxx	Enable high current input (HCI) mode for input channel xx of group 2.
Signal outputs	IN_Chxx	Indicates the state of input signal from channel xx.
Control outputs	Over_Current	Over current channel number
	ERROR	Error Code

3.1.2 i201_LED_Config

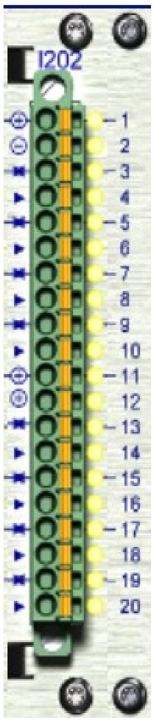
Name	**i201_IO_LED_Config**	
Purpose	Sets the LEDs into user controlled mode or input controlled mode	
Module Class	i201	
Symbol		
IEC-61131 Type	Function	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	isUser	TRUE: Sets the LED to be controlled by i201_LED_Set function FALSE: Sets the LES to be controlled by the state of its corresponding channel.
Output	ERROR	Error Code

3.1.3 i201_LED_Set

Name	**i201_LED_Set**	
Purpose	Set the LEDs ON or OFF	
Module Class	i201	
Symbol		
IEC-61131 Type	Function	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	LED_Ch01..16	TRUE: Sets the LED ON FALSE: Sets the LED OFF.
Output	ERROR	Error Code

3.2 i202 Digital Input / Output Module – i202_DG_FW_LIB Library

i202 Hardware interface



Pin	Channel	Group	Meaning
1	-		Vbat+
2	-		Vbat-
3	1	1	IN / OUT
4	2	2	IN
5	3	1	IN / OUT
6	4	2	IN
7	5	1	IN / OUT
8	6	2	IN
9	7	1	IN / OUT
10	8	2	IN
11	-		Vbat+
12	-		Vbat-
13	9	1	IN / OUT
14	10	2	IN
15	11	1	IN / OUT
16	12	2	IN
17	13	1	IN / OUT
18	14	2	IN
19	15	1	IN / OUT
20	16	2	IN

i202 Software Interface

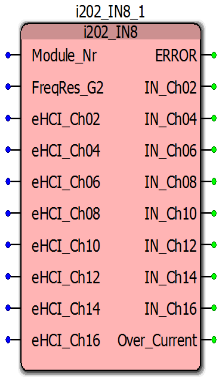
The following function blocks are included in the i202_DG_FW_LIB library

i202_IN8	Get digital input signals (normal inputs)	input
i202_IN8_RB	Get digital input signals (read-back inputs)	input
i202_LED_Set	Set the LEDs ON or OFF	indication
i202_IO_LED_Config	Sets the LEDs into user controlled mode or input controlled mode	control
i202_OUT8_Cycle	Set the duty cycle in percent of the pulse width for an output channel, when it is configured as a PWM output. [resolution 1 %]	PWM output
i202_OUT8_Pulse	Set the pulse width (in microseconds) for the corresponding output channel, when it is configured as a PWM output. [resolution 1 microsecond]	PWM output
i202_OUT8_State	Set the state of an output channel, when it is configured as a state output.	state output
i202_Set_OUT_Type	Select channels to be used as PWM outputs	control

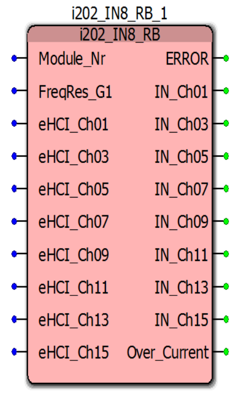
When selecting function blocks for use on the same I/O module please consider the combination strategy indicated by the last columns and the following key:

	Forbidden combination
	Mandatory combination
	Recommended combination

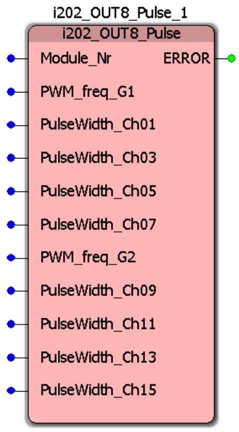
3.2.1 i202_IN8

Name	i202_IN8	
Purpose	Get digital input signals (normal inputs)	
Module Class	i202	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	eHCI_Chxx	Enable high current input (HCI) mode for input channel xx of group 2.
Signal outputs	IN_Chxx	Indicates the state of input signal from channel xx.
Control outputs	Over_Current	Over current channel number
	ERROR	Error Code

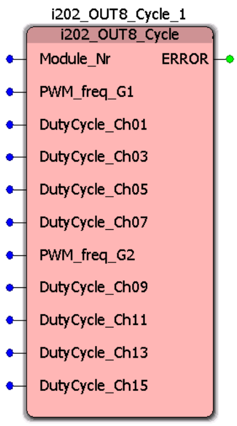
3.2.2 i202_IN8_RB

Name	i202_IN8_RB	
Purpose	Get digital input signals (read-back inputs)	
Module Class	i202	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	eHCI_Chxx	Enable high current input (HCI) mode for inputs channel xx of group 1.
Signal outputs	IN_Chxx	Indicates the state of I/O signal from channel xx.
Control outputs	Over_Current	Over current channel number
	ERROR	Error Code

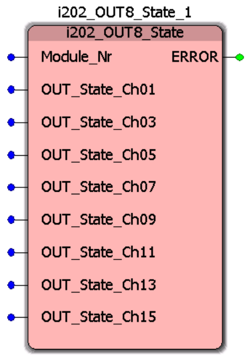
3.2.3 i202_OUT8_Pulse

Name	i202_OUT8_Pulse	
Purpose	Set the pulse width (in microseconds) for the corresponding output channel, when it is configured as a PWM output. [resolution 1 microsecond]	
Module Class	i202	
Symbo		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
Inputs	PWM_ freq_G1	Base PWM Frequency of Group 1. 0 = PWM disabled (default) Frequency Range: 25 … 10000Hz
	PulseWidth_Ch01,3,5,7	Pulse width in microseconds. 0 = PWM Channel disabled (default)
	PWM_ freq_G2	Base PWM Frequency of Group 2. 0 = PWM disabled (default) Frequency Range: 25 … 10000Hz
Output	ERROR	Error Code
Remark	Requires i202_Set_OUT_Type with Set_Type_Pulse_Chnn set to TRUE for each channel nn used as a PWM output.	

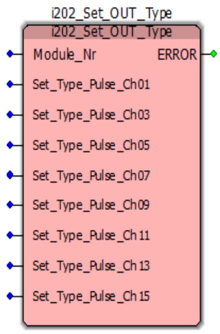
3.2.4 i202_OUT8_Cycle

Name	**i202_OUT8_Cycle**	
Purpose	Set the duty cycle in percent of the pulse width for an output channel, when it is configured as a PWM output. [resolution 1 %]	
Module Class	i202	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	PWM_ freq_G1	Base PWM Frequency of Group 1. 0 = PWM disabled (default) Frequency Range: 25 … 10000Hz
	DutyCycle_Ch01,3,5,7	Duty Cycle in % [range 1 … 100] 0 = PWM Channel disabled (default)
	PWM_ freq_G2	Base PWM Frequency of Group 2. 0 = PWM disabled (default) Frequency Range: 25 … 10000Hz
	DutyCycle_Ch02,4,6,8	Duty Cycle in % [range 1 … 100] 0 = PWM Channel disabled (default)
Output	ERROR	Error Code
Remark	Requires i202_Set_OUT_Type with Set_Type_Pulse_Chnn set to TRUE for each channel nn used as a PWM output.	

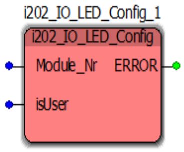
3.2.5 i202_OUT8_State

Name	**i202_OUT8_State**	
Purpose	Set the state of an output channel, when it is configured as a state output.	
Module Class	i202	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
Inputs	OUT_State_Ch01,3,5, 7,9,11,13,15	Channel outputs set HIGH when TRUE and set LOW when FALSE
Output	ERROR	Error Code
Remark	Requires i202_Set_OUT_Type with Set_Type_Pulse_Chnn set to FALSE for each channel nn used as a state output. The Module_Nr numbers must match.	

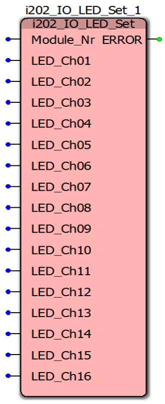
3.2.6 i202_Set_OUT_Type

Name	**i202_Set_OUT_Type**	
Purpose	Select channels to be used as PWM outputs	
Module Class	i202	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr	Indicates which I/O module in the ionia rack is addressed.
Inputs	Set_Type_Pulse_Ch01,3, 5,7,9,11,13,15	Channel outputs PWM when TRUE and state when FALSE
Output	ERROR	Error Code
Remark	The function block (FB) is designed to be used in combination with state and PWM blocks to select the output type. This FB controls FBs with matching Module_Nr numbers.	

3.2.7 i202_LED_Config

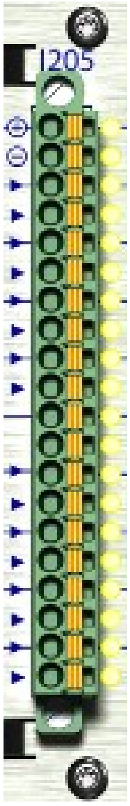
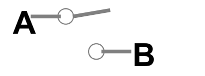
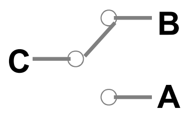
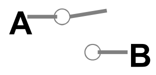
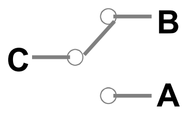
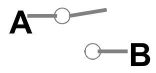
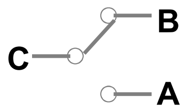
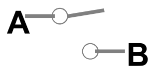
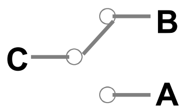
Name	i202_IO_LED_Config	
Purpose	Sets the LEDs into user controlled mode or input controlled mode	
Module Class	i202	
Symbol		
IEC-61131 Type	Function	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	isUser	TRUE: Sets the LED to be controlled by i202_LED_Set function FALSE: Sets the LES to be controlled by the state of its corresponding channel.
Output	ERROR	Error Code

3.2.8 i202_LED_Set

Name	i202_LED_Set	
Purpose	Set the LEDs ON or OFF	
Module Class	i202	
Symbol		
IEC-61131 Type	Function	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	LED_Ch01..16	TRUE: Sets the LED ON FALSE: Sets the LED OFF.
Output	ERROR	Error Code

3.3 i205 Relay Output Module – i205_DG_FW_LIB Library

i205 Hardware Interface

	Pin	Chan	Usr LED	Relay Pin	Relay Open State
	1	1		A	
	2	1	1	B	
	3	2	2	B	
	4	2		C	
	5	2	3	A	
	6	3		A	
	7	3	4	B	
	8	4	5	B	
	9	4		C	
	10	4	6	A	
	11	5		A	
	12	5	7	B	
	13	6	8	B	
	14	6		C	
	15	6	9	A	
	16	7		A	
	17	7	10	B	
	18	8	11	B	
	19	8		C	
	20	8	12	A	

i205 Software Interface

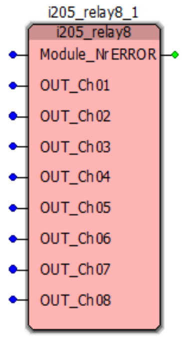
The following function blocks are included in the i205_DG_FW_LIB library

i205_relay8	Set relay output channel	control				
i205_readback8	Readback state of relay output	control				
i205_IO_LED_Config	Sets the LEDs into user controlled mode or input controlled mode	control				
i205_IO_LED_Set	Set the LEDs ON or OFF	indication				

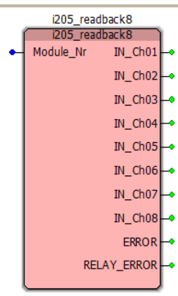
When selecting function blocks for use on the same I/O module please consider the combination strategy indicated by the last columns and the following key:

	Forbidden combination
	Mandatory combination
	Recommended combination

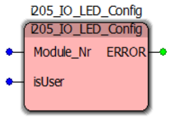
3.3.1 i205_relay8

Name	**i205_relay8**	
Purpose	Set relay output channel	
Module Class	i205	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
Inputs	OUT_Ch01..08	Channel outputs set HIGH when TRUE and set LOW when FALSE
Output	ERROR	Error Code

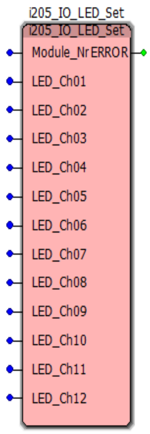
3.3.2 i205_readback8

Name	**i205_readback8**	
Purpose	Readback the state of the output channel	
Module Class	i205	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
Inputs	IN_Ch01..08	State ouf the output channel
Output	ERROR	Error Code

3.3.3 i205_LED_Config

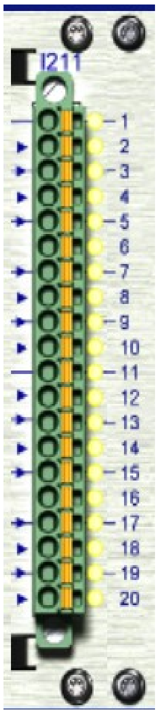
Name	**i205_LED_Config**	
Purpose	Sets the LEDs into user controlled mode or input controlled mode	
Module Class	i205	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	isUser	0: Input enable controlled by FB i205_relay8 0: User LED. Controlled by FB i205_IO_LED_Set.
Output	ERROR	DG_Error Code.

3.3.4 i205_LED_Set

Name	**i205_LED_Set**	
Purpose	Sets the LEDs when in user mode.	
Module Class	i205	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	Gate_Nr.	Gate 1...4
Input	LED_Ch01 … LED_Ch12	0: LED off 1: LED on
Output	ERROR	DG_ERROR Code -1 : Gate number not plausible. (valid values 1...4)

3.4 i211 Analog Input – i211_DG_FW_LIB library

i211 Hardware Interface



Pin	Chan	Group
1	-	-
2	1	G1
3	2	G1
4	3	G1
5	4	G1
6	-	-
7	5	G2
8	6	G2
9	7	G2
10	8	G2
11	-	-
12	9	G3
13	10	G3
14	11	G3
15	12	G3
16	-	-
17	13	G4
18	14	G4
19	15	G4
20	16	G4

i211 Software Interface

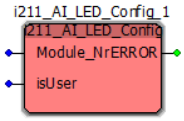
The following function blocks are included in the i211_DG_FW_LIB library

i211_LED_Config	Sets the LEDs into user controlled mode or input controlled mode	control
i211_LED_Set	Sets the LEDs when in user mode	output
i211_AIN_V	Voltage Measurement	input
i211_AIN_C	Current Measurement	input
i211_AIN_R	Resistive Measurement	input

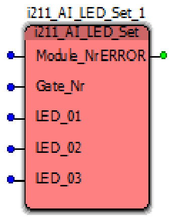
When selecting function blocks for use on the same I/O module please consider the combination strategy indicated by the last columns and the following key:

	Forbidden combination
	Mandatory combination
	Recommended combination

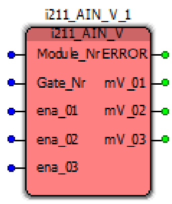
3.4.1 i211_LED_Config

Name	i211_LED_Config	
Purpose	Sets the LEDs into user controlled mode or input controlled mode	
Module Class	i211	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	isUser	0: Input enable controlled by FB i211_AIN_V,C,R,P 0: User LED. Controlled by FB i211_AI_LED_SET.
Output	ERROR	DG_Error Code.

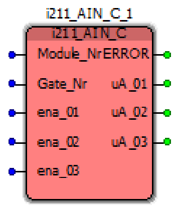
3.4.2 i211_LED_Set

Name	i211_LED_Set	
Purpose	Sets the LEDs when in user mode	
Module Class	i211	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	Gate_Nr.	Gate 1...4
Input	LED_01 … LED_03	0: LED off 1: LED on
Output	ERROR	DG_ERROR Code -1 : Gate number not plausible. (valid values 1...4)

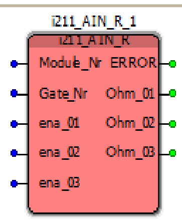
3.4.3 i211_AIN_V

Name	i211_AIN_V	
Purpose	Voltage Measurement	
Module Class	i211	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	Gate_Nr.	Gate 1...4
Input	ena_01 … ena_03	0: Measurement on channel mV_01..3 deactivated. 1: Measurement on channel mV_01..3 activated.
Output	mV_01 … mV_03	Process value in mV.

3.4.4 i211_AIN_C

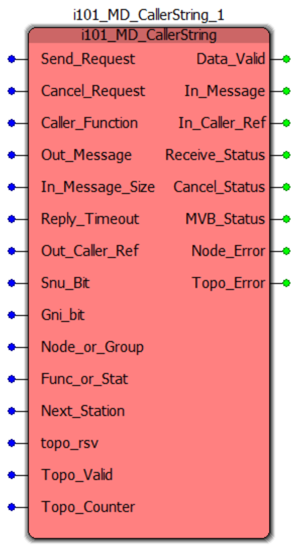
Name	i211_AIN_C	
Purpose	Current Measurement	
Module Class	i211	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	Gate_Nr.	Gate 1...4
Input	ena_01 … ena_03	0: Measurement on channel uA_01..3 deactivated. 1: Measurement on channel uA_01..3 activated.
Output	uA_01 … uA_03	Process value in uA.

3.4.5 i211_AIN_R

Name	i211_AIN_R	
Purpose	Resistive Measurement	
Module Class	i211	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Module_Nr.	Indicates which I/O module in the ionia rack is addressed.
	Gate_Nr.	Gate 1...4,
Input	ena_01 … ena_03	0: Measurement on channel Ohm_01..3 deactivated. 1: Measurement on channel Ohm_01..3 activated.
Output	Ohm_01 … Ohm_03	Process value in Ohm.

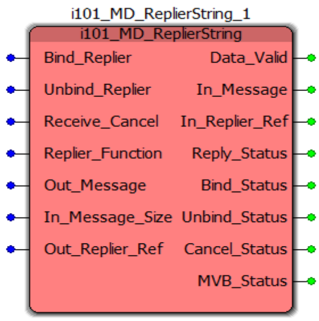
3.5 MVB Message Data Communication – i101_DG_MVB_LIB Library

3.5.1 i101_MD_CallerString

Name	**i101_MD_CallerString**	
Purpose	Send a string to a replyer and receive a confirmation string of the replyer.	
Module Class	i101	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Send_Request	FALSE : Functionblock in disabled. TRUE : Call request is emitted.
	Cancel_Request	Do not use.
	Caller_Function	Function ID of the caller.
	Out_Message	String to be sent to the replier.
	In_Message_Size	Size of the expected receive string in Bytes.
	Reply_Timeout	Timeout value in multiples of 64 ms for the reply after the transfer of the Call_Message.
	Out_Caller_Ref	External reference for the call to be returned by the replier. It can be used in any way by the caller.
	Continued on next page!	

Name	i101_MD_CallerString	
Reply Address	Application address of the Replier function or station. Tpye of AM_ADDRESS.	
	Snu_Bit	System not user bit. FALSE : Indicates a User_Address TRUE : Message uses system addressing.
	Gni_Bit	Group Not Individual. FALSE : Indicates an individual (NODE) address. TRUE : Indicates a Group Address
	Node_or_Group	Gni_Bit = FALSE : Specifiy a Node Address (6 Bit) Gni_Bit = TRUE : Specify a Group Address (6 Bit)
	Func_or_Stat	Snu_Bit = FALSE : Specify a Function ID Snu_bit = TRUE : Specify a Station ID
	Next_Station	Next Stadion ID
	Topo_Rsv	Reserved, always FALSE.
	Topo_Valid	Inidcates if the following Topo_Counter is valid.
	Topo_Counter	Topo_Counter or AM_ANY_TOPO (=0).
Output Data	Data_Valid	FALSE : Received message is invalid resp. old. TRUE : Received message is valid.
	In_Message	Replied Message (String) from the replier.
	In_Caller_Ref	Returned external reference from the replier.
	Receive_Status	Type : AM_RESULT Default value : AM_FAILURE
	Cancel_Status	Do not use.
	MVB_Status	Type : I101_ERROR_CODE Default value : AM_FAILURE
	Node_Error	FALSE : Input Node_or_Group within range of 6Bit TRUE : Input Node_or_Group outside range of 6Bit
	Topo_Error	FALSE : Input Topo_Counter within range of 6Bit TRUE : Input Topo_Counter outside range of 6Bit

3.5.2 i101_MD_ReplierString

Name	**i101_MD_ReplierString**	
Purpose	Receive a string from the caller and confirm the message with a reply string.	
Module Class	i101	
Symbol		
IEC-61131 Type	Function Block	
Parameters	Bind_Replier	FALSE : Functionblock in disabled. TRUE : Makes a Replier known to the Messenger and connects its procedures.
	Unbind_Replier	Do not use.
	Receive_Cancel	Do not use.
	Replier_Function	Function_Id of the Replier expecting a call. Function_Id = 253 implies a System_Address.
	Out_Message	String to be sent to the caller.
	In_Message_Size	Size of the expected caller string in Bytes.
	Out_Replier_Ref	External reference returned with the related receive_confirm procedure. It is at the same time an instance reference and distinguishes the Replier instances which serve the same Replier.
	Continued on next page!	

Name	i101_MD_ReplierString	
Output Data	Data_Valid	FALSE : Received message is invalid resp. old. TRUE : Received message is valid.
	In_Message	Called Message (String) from the caller.
	In_Replier_Ref	Returned external reference from the caller.
	Reply_Status	Type : AM_RESULT Default value : AM_FAILURE
	Bind_Status	Type : AM_RESULT Default value : AM_FAILURE
	Unbind_Status	Do not use.
	Cancel_Status	Do not use.
	MVB_Status	Type : I101_ERROR_CODE Default value : AM_FAILURE

4. Data Types and Constants

4.1 AM_RESULT

```
AM_OK                  = 0     /* successful termination                    */
AM_FAILURE             = 1     /* unspecified failure                       */
AM_BUS_ERR             = 2     /* no bus transmission possible              */
AM_REM_CONN_OVF        = 3     /* too many incoming connections             */
AM_CONN_TMO_ERR        = 4     /* Connect_Request not answered              */
AM_SEND_TMO_ERR        = 5     /* send time-out (connect was OK)            */
AM_REPLY_TMO_ERR       = 6     /* no reply received                         */
AM_ALIVE_TMO_ERR       = 7     /* no complete message received              */
AM_NO_LOC_MEM_ERR      = 8     /* not enough memory or timers               */
AM_NO_REM_MEM_ERR      = 9     /* no more memory or timer (partner)         */
AM_REM_CANC_ERR        = 10    /* cancelled by partner                      */
AM_ALREADY_USED        = 11    /* same operation already done               */
AM_ADDR_FMT_ERR        = 12    /* address format error                      */
AM_NO_REPLY_EXP_ERR    = 13    /* no such reply expected                    */
AM_NR_OF_CALLS_OVF     = 14    /* too many calls requested                  */
AM_REPLY_LEN_OVF       = 15    /* Reply_Message too long                    */
AM_DUPL_LINK_ERR       = 16    /* duplicated conversation error             */
AM_MY_DEV_UNKNOWN_ERR  = 17    /* my device unknown or not valid            */
AM_NO_READY_INST_ERR   = 18    /* no ready Replier instance                 */
AM_NR_OF_INST_OVF      = 19    /* too many Replier instances                */
AM_CALL_LEN_OVF        = 20    /* Call_Message too long                     */
AM_UNKNOWN_DEST_ERR    = 21    /* partner device unknown                    */
AM_INAUG_ERR           = 22    /* train inauguration occurred               */
AM_TRY_LATER_ERR       = 23    /* (internally used only)                    */
AM_FIN_NOT_REG_ERR     = 24    /* final not registered                      */
AM_GW_FIN_NOT_REG_ERR  = 25    /* final not registered in router            */
AM_GW_ORI_NOT_REG_ERR  = 26    /* origin not registered in router           */
AM_MAX_ERR             = 31    /* highest system code                       */
                               /* user codes are higher than 31      */
```

4.2 DG_ERROR Code

```
DG_OK                               0  /* No error                              */
DG_ERROR                          400  /* General host driver error             */
DG_ERROR_TIMEOUT                  401  /* Timeout on read/write acess           */
DG_ERROR_COMM_FAIL                402  /* Comm.failure                          */
DG_ERROR_OSL_SYNC_INIT            500  /* Mutex initialisation failed           */
DG_ERROR_OSL_SYNC_LOCK            501  /* Mutex lock failed                     */
DG_ERROR_OSL_SYNC_UNLOCK          502  /* Mutex unlock failed                   */
DG_ERROR_OSL_SYNC_DESTROY         503  /* Mutex destroy failed                  */
DG_ERROR_OSL_INIT                 520  /* I/O access initialisation failed      */
DG_ERROR_OSL_READ                 521  /* Read access failed                    */
DG_ERROR_OSL_WRITE                522  /* Write access failed                   */
DG_ERROR_OSL_NOT_READY            523  /* device registers not ready            */
DG_ERROR_HDIO_RECV_WOULD_BLOCK    541  /* Receiving task would block            */
DG_ERROR_HDIO_SEND_WOULD_BLOCK    542  /* Sending task wouldblock               */
DG_ERROR_HDIO_UNKNOWN_CHANNEL_TYPE 544 /* Unknown channel type                  */
DG_ERROR_HDIO_UNKNOWN_IO_MODE     545  /* I/O mode unknown                      */
DG_ERROR_HDIO_NOT_SUPPORTED       546  /* Not supported forconfigured I/O mode  */
DG_ERROR_HDC_PROTOCOL_MISMATCH    560  /* Received wrong protocol ID            */
DG_ERROR_HDC_PROTOCOL             561  /* Error in HDC protocol                 */
DG_ERROR_HDC_HANDLE               562  /* Handle is wrong                       */
DG_ERROR_HDC_LENGTH_MISSMATCH     563  /* Payload sz. Spec. in header not avail. */
DG_ERROR_RPC_UNKNOWN_TYPE         580  /* Unknown RPC type                      */
DG_ERROR_VERSION_DRIVER_OLD       590  /* host needs a newer device             */
DG_ERROR_VERSION_DEVICE_OLD       591  /* device requires a newer host          */
DG_ERROR_EMPTY_BLOCK              592  /* protocol not ready                    */
DG_ERROR_REGISTER_NOT_CORRECT     593  /* norm. means the device has been reset */
DG_ERROR_FUNCTION_TIMED_OUT       594  /* time to execute function exceeds limit */
DG_ERROR_CHANNEL_TIMED_OUT        595  /* previous function executed on this
                                          channel exceeded timout limit         */
```

4.3 Address Constants

```
AM_SAME_STATION        = 0     /* This station, regardless of Station ID    */
AM_UNKNOWN             = 255   /* Unknown Station ID                        */
AM_MAX_BUSSES          = 1..16 /* Maximum number of link layers supportet   */
                               /* by an  implemntation                      */
AM_ROUTER_FCT          = 254   /* Function ID of Router                     */
AM_AGENT_FCT           = 253   /* Function ID of Agent                      */
AM_MANAGER_FCT         = 254   /* Function ID of Manager                    */
AM_SAME_NODE           = 0     /* Communication under the same Node, not    */
                               /* going over the train bus                  */
AM_SYSTEM_ADDR         = 128   /* Bit 0 of the Node Address indicates a     */
                               /* System Address                            */
AM_ANY_TOPO            = 0     /* Top Counter in unknown                    */
```

4.4 101_ERROR_CODE

```
I101_OK                = 0     /* correct termination                       */
I101_DEVICENAME        = 1     /* Device Name plausibility Error            */
I101_DEVICEADDRESS     = 2     /* Device Address plausibility Error         */
I101_DEVICEBAENABLE    = 3     /* Bus admininstrator plausibility Error     */
I101_RESOURCENAME      = 4     /* Resource Name plausibility Error          */
I101_APPDOMAIN         = 10    /* Could not open application domain         */
```

```
I101_APPFILECREATE      = 11    /* Could not create fd in app domain          */
I101_APPFILEREAD        = 12    /* COULD NOT READ FILE FROM APP DOMAIN        */
I101_FILEOPEN           = 13    /* COULD NOT CREATE FILE HANDLE               */
I101_FILEWRITE          = 14    /* COULD NOT WRITE FILE                       */
I101_AP_RESULT          = 20
I101_AS_RESULT          = 21
I101_CM_RESULT          = 22
I101_MM_RESULT          = 23
I101_PVREAD             = 30    /* COULD NOT READ PV_NAME                     */
I101_PVWRITE            = 31    /* COULD NOT WRITE PV_NAME                    */
I101_UNSPECIFIED        = 90    /* Unspecified error                          */
```

5. Document History

d-006935-014140

- The description of the FW library i20x_DG_FW_LIB is splited off into independent chapters for each FW library; i.e. i201DG_FW_LIB , i202 DG_FW_LIB.
- Function blocks for library i201_DG_FW_LIB added.
- Function blocks for library i205_DG_FW_LIB added.
- Function blocks for library i101_DG_MVB_LIB added; i.e. MVB Message Data.
- CANopen Driver i103_DG_CO_DRV and its Driver Dialog i103_DG_CO_DLG added to chapter Revision Information.
- DG_ERROR code added.

d-006935-013206

- Initial version for G2.1 release.

6. Index

Appendix A: Document Numbering System

All Duagon documents have a unique identification number. The identification number has a certain internal structure in order to ease the tracking of different documents. In general, there are two parts:

Prefix	Document number	Filing number
d	-000310	-001952
Always constant	Specifies a certain purpose of a document with the intention to link several documents with different filing number. Please note, that the purpose of the document number is not stored for each document number, but can be derived from the document title, which is stored for each Filing number. The format is either 6 digits or not available.	Unique number, that identifies a particular document. Released in sequential manner as the documents are filed in the archive. A Duagon- internal data base contains exactly one document title text for each filing number. Always 6 digits.

Examples for identification numbers

Identification number	Document Title / Remarks
d-000310-001606 d-000310-001952	„DXIO data sheet Rev 2.2" „DXIO data sheet Rev 2.3" A document, that is updated from time to time: the document number has the purpose to link several versions of the „DXIO data sheet" together. The filing number distinguishes between different versions. Please note, that the document number part is kept the same, as long as the basic intention of the early versions is still kept, for example during revisions due to debugging or manufacturing updates. In case a significant change happens, another document number would be applied.
d-000719	„Notes from prototype meeting ..." A document, that is obviously not updated after release. The „document number" part is missing and the filing number remains the only used part for identification.

Recommendation:
In your order, you may specify for example "d-000584-nnnnnn" in order to get the "newest" version of a specific product. When you do not want to follow the sequence of newer versions, i.e. you want to stick to a specific version, then specify the full identification number like "d-000584-002043".